Filtration Solutions to Algae Growth in Treatment Ponds

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Introduction

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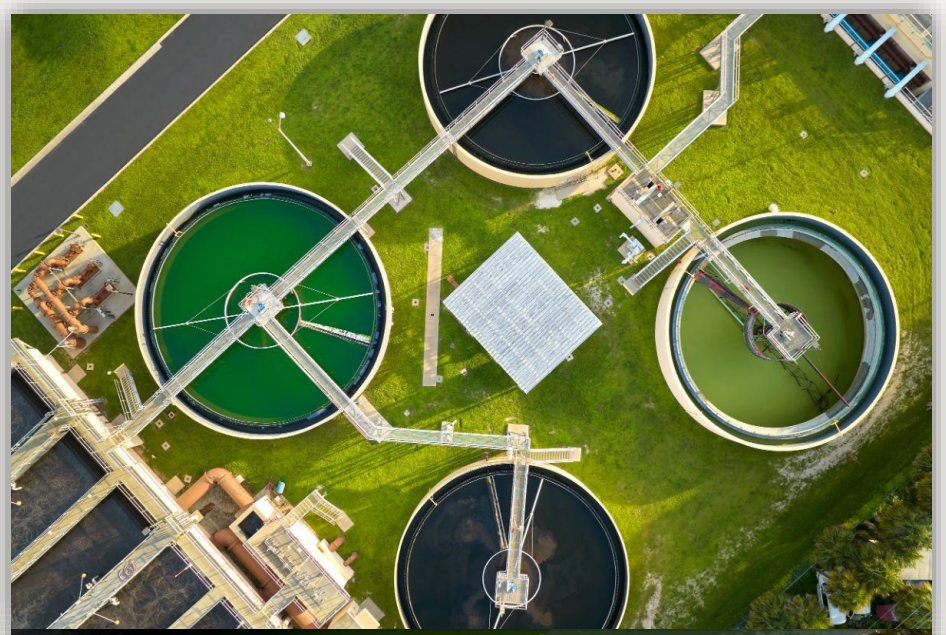
Scenario

- Nutrient richness leads to cyanotoxins
- Reduces overall efficiency
- Tampakan, Philippines equatorial climate
- Hepatotoxins such as microcystin have negative health effects on livestock



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Problem statement

Design a hollow fiber membrane water filter to prevent toxic blue-green algae proliferation in treatments ponds at a plant in Tampakan, Philippines. This system should be effective, remain cost-efficient for large scale deployment, and minimize environmental impact through sustainable materials and low energy consumption.

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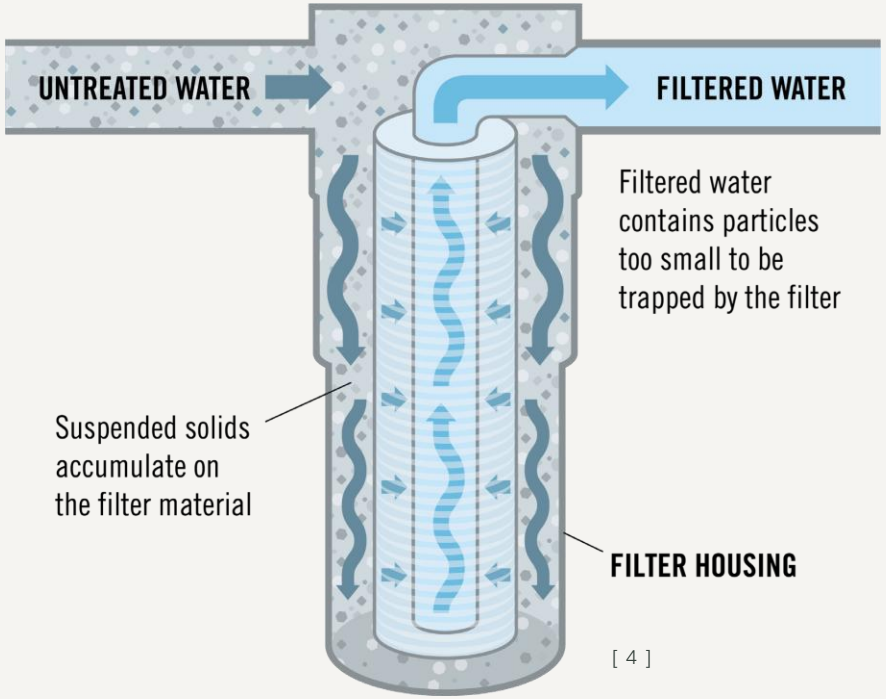
What is mechanical filtration?

- The removal of contaminating particles by passing a fluid through a filtration medium
- The filtration medium traps the particles and allows the purified fluid to continue
- The medium consists of a porous material, trapping any particles larger than the pores
- This technique is effective only for the removal of solids
- Alternative methods: reverse osmosis
  - May have been effective for algae removal
  - Risks filter longevity and high operating cost



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Key Objectives & Metrics

- **Objective 1: Effective Treatment**

Metric: % reduction of cyanobacteria biomass (μg/L)

- **Objective 2: Environmentally Responsible**

Metric: kg CO_2 per m^3 treated water

- **Objective 3: Cost Effective**

Metric: Cost per filter unit ($)

Lower CO_2 + Lower cost + Higher removal efficiency = Better performance

Regulations

Contaminant and Waste Standards:

- DENR regulates bacterial/chemical concentration in biomass per unit volume of water treated.
  → Hepatotoxin microcystin is regulated to a maximum of 1μg/L.
- pH of discharged water is standardized between 6.5 - 9.0 to protect aquatic ecosystems.

Filter Material Acquisition:

- Filter materials must be non-toxic and must not contribute additional pollutants.
- National water quality policy requires filter materials to not be the cause of secondary contamination

Sustainability considerations/practices

- **Material Durability**

It does not degrade early on in its lifetime

- **Reuse and Maintenance based sustainability**

It should be back washable

- **Energy Consciousness & Standards**

It should require less energy to filter out the same amount of substances

- **Environmental Protection through Effective treatment**

By effectively filtering the water, we are ensuring no harmful toxins are entering the water stream hence benefiting the environment

Material Selection Process

Candidate Materials:

- Sisal Fiber
- Coir Fiber
- Ramie Fiber

Simple Decision Matrix - Template			
	Material 1: Sisal Fiber	*Material 2: Coir Fiber*	*Material 3: Ramie Fiber*
Ease of maintenance	3	5	4
Ease of access to material	3	2	1
Hydrophilicity (moisture regain)	4	3	5
Individual filter fiber thickness	4	3	5
TOTAL	14	13	15

Material Selection Methods:

- Ashby Material selection using ANSYS Granta EduPack
- Material Performance Index (MPI)
- Decision Matrix

Ramie Fiber

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Coir Fiber

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Sisal Fiber

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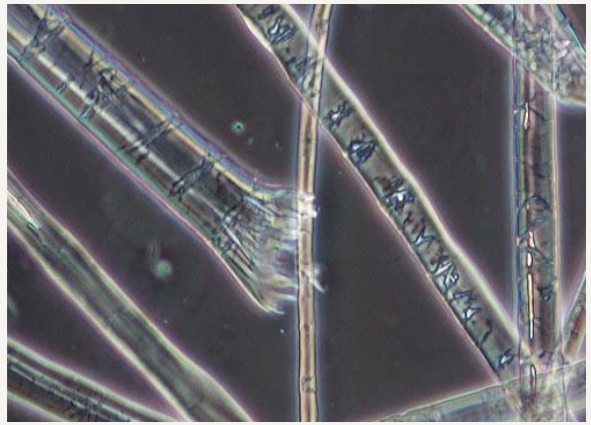
Final Material Choice: Ramie Fiber



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Key Mechanical Properties

- Young's Modulus: **86 GPa**
- Yield Strength: **531 MPa**
- Density: **1500 kg/m^3**

- Why Ramie Fiber?
- ✓ Thin fibers (25–30 µm)
  → Higher fiber density improves filtration efficiency
- ✓ High hydrophilicity
  → Better interaction with water and algae particles
- ✓ Strong mechanical performance
  → Maintains stiffness and strength under filtration loads
- ✓ Environmentally sustainable natural fiber

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Porosity Choice and Explanation

Contaminant Radius: 1 – 100 μm

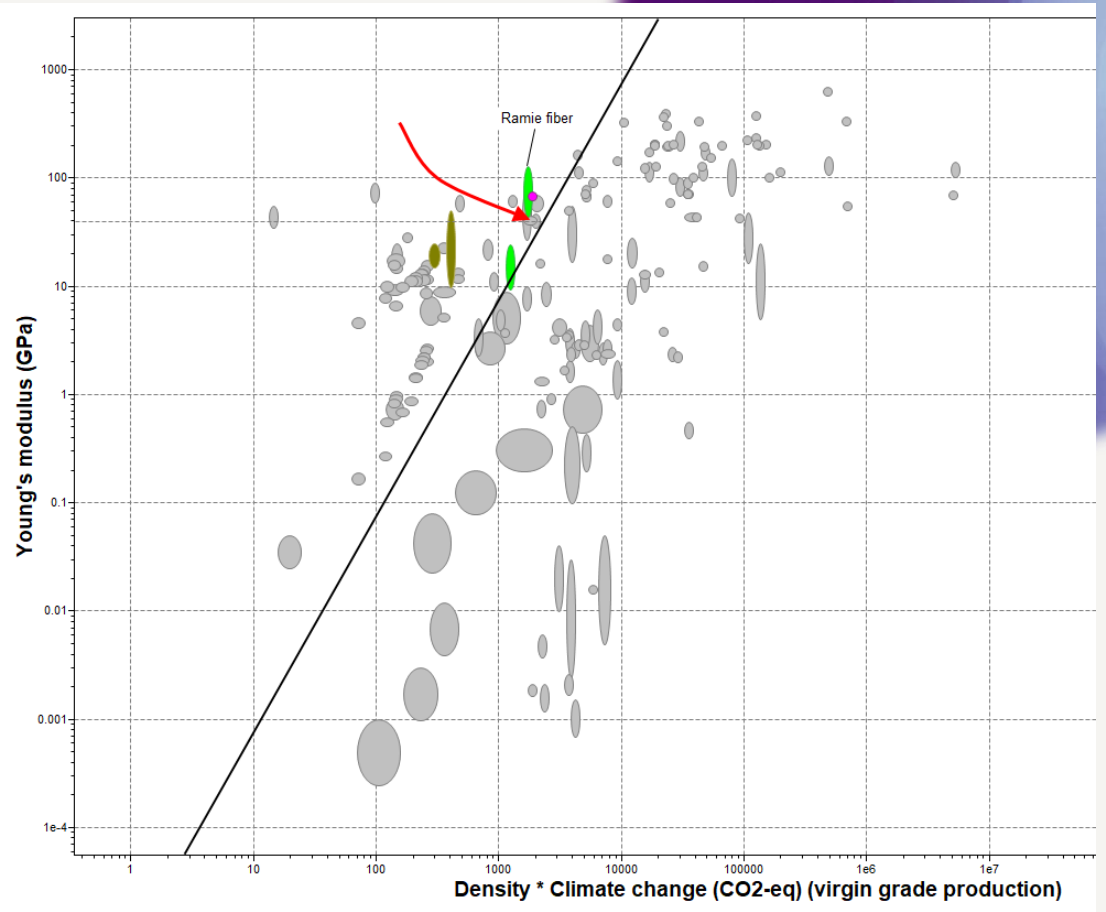
Final Porosity: 0.305

Calculation Choices:

- **Micropore Radius:** 0.75 μm
- **Number of Micro Pores:** 2,000,000
  - Surface area of filter fiber/micro pore area

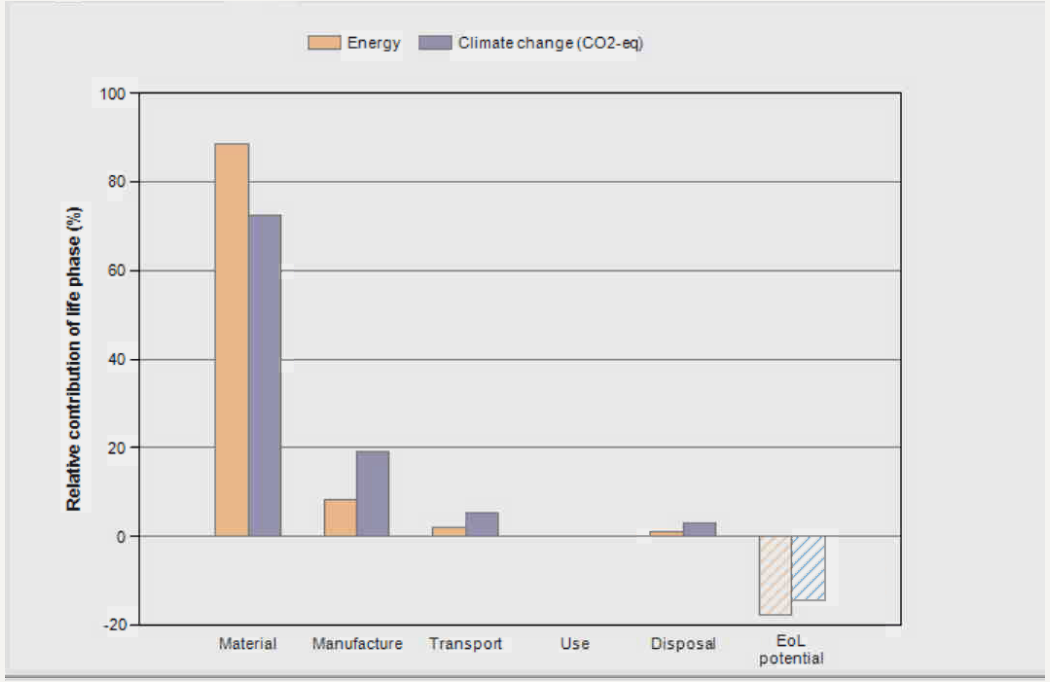
$E_{effective}$: 41.6 GPa

$\sigma_{effective}$: 257 MPa



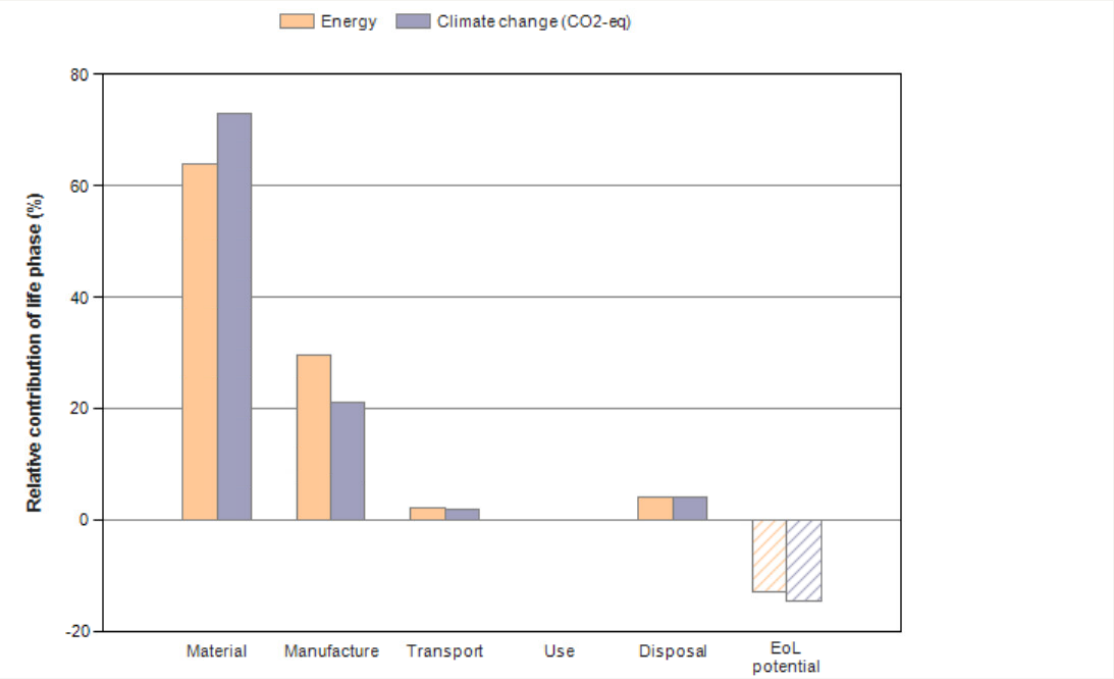
Eco Audits

Ramie Fiber



- Significant Energy use in the material phase
- Very little usage in the manufacture and transport phase

Coir Fiber



- Significant Energy use in the material phase
- Moderate usage in the manufacturing phase
- Very little usage in the transport phase

Conclusion

-Final Material Choice: Ramie Fiber

-“Each filter requires 9.214 g of fiber, costing approximately $0.09in filter fibers per filter

-Recyclable

-Pore radius of 0.75, smaller than contaminants

[3]

Questions

Citations

- For photos use in slides

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